Python (part1)

Installing Anaconda

Link to download Anaconda

https://www.anaconda.com/distribution/#windows

Steps to download Anaconda:

https://docs.anaconda.com/anaconda/install/windows/

Exploring Anaconda

To be illustrated in the lecture.

Common Libraries

NumPy

It has advanced math functions, scientific computing package and useful features for operations on n-arrays and matrices in Python.

SciPy

SciPy is a library of software for engineering and science. SciPy contains modules for linear algebra, optimization, integration, and statistics. The main functionality of SciPy library is built upon NumPy, and thus, its arrays make substantial use of NumPy.

Pandas

It designed for quick and easy data manipulation, aggregation, and visualization.

Matplotlib

For visualization

Seaborn
For advanced visualization

SciKit-Learn
Designed for specific functionalities like image processing and machine learning facilitation.

Theano, **TensorFlow**, and **Keras**：are used for Deep Learning**.**

NLTK and **Gensim**: are used for Natural Language Processing.

Selecting dataset

Iris.csv

Some basic operations over the dataset

Load The Data

We are using pandas to load the data. We will also use pandas next to explore the data both with descriptive statistics and data visualization.

```
import pandas

dataset = pandas.read_csv("C:\\Users\\USER\\Downloads\\iris_dataset.csv")
```

Dimensions of Dataset

```
print(dataset.shape)
```

Top rows of the Data

```
print(dataset.head(5))
```

Statistical Summary

```
print(dataset.describe())
```

Class Distribution

```
print(dataset.groupby('variety').size())
```

Create a Validation Dataset

```
from sklearn import model_selection
array = dataset.values
X = array[:,0:4]
Y = array[:,4]
validation_size = 0.20
seed = 7
X_train, X_validation, Y_train, Y_validation = model_selection.train_test_split(X, Y,
test_size=validation_size, random_state=seed)
```

Make Predictions

```
from sklearn.neighbors import KNeighborsClassifier
knn = KNeighborsClassifier()
knn.fit(X_train, Y_train)
predictions = knn.predict(X_validation)
from sklearn.metrics import accuracy_score
print(accuracy_score(Y_validation, predictions))
from sklearn.metrics import confusion_matrix
print(confusion_matrix(Y_validation, predictions))
```